PREM KUMAR R

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Education

Government College Of Engineering Bargur **November 2022 - April 2026**
B.E Computer Science and Engineering (CGPA-7.66) *Bargur, Tamil Nadu*

Experience

Infosys Springboard Internship **Nov 2025 - Jan 2026**
Toonify – AI Cartoonization Web App | *Python, Streamlit, OpenCV, PIL, SQLite* **Hybrid, India**

- Developed a full-stack AI-powered web application to convert real-world images into cartoon-style artwork using computer vision techniques.
- Implemented secure authentication and role-based access (user/admin) with session management using Python, Streamlit, and SQLite.
- Designed and optimized image cartoonization pipelines using OpenCV and Python Imaging Library, improving processing efficiency and output quality.
- Built a simulated payment gateway with transaction tracking and an admin dashboard for user and analytics management.
- Followed structured development practices using SDLC and Agile Workflow.

Projects

Prompt Based Image Manipulation Using Diffusion Model | *Python, PyTorch, CLIP, Computer Vision* **Nov 2025**

- Implemented a deep learning-based image editing system to modify real images using natural language prompts.
- Utilized pre-trained Stable Diffusion models for text-guided image generation and editing.
- Explored text-image alignment using CLIP-based similarity to ensure semantic consistency.
- Conducted experiments with various prompts and images to evaluate editing quality and model behavior.

Audio Deepfake Detection **April 2026**

- Developed a deep learning-based audio deepfake detection system to distinguish between real and AI-generated speech.
- Implemented audio feature extraction techniques such as MFCC, spectrograms, and temporal features for effective model training.
- Trained and evaluated machine learning and neural network models to improve classification accuracy and robustness against manipulated audio samples.

Technical Skills

Languages: Python, C, CPP
Developer Tools: VS Code, Google Colab

Certifications

- **Google:** Google Cloud Engineering Certificate
- **Google:** Google Cloud Data Analytics Certificate
- **Coursera:** Programming For Everybody
- **Infosys:** Python Foundation Certification
- **Guvi:** Generative AI
- **Guvi:** Python

Languages

Tamil: Native

English: Proficient